Md. Ashik-Uz-Zaman

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Education

East West University — Fall 2023 – Present
B.Sc. in Computer Science and Engineering — *CGPA: 3.60 / 4.00 (117 credits completed)*

Dhaka Imperial College — 2021 – 2022
Higher Secondary Certificate (Science) — *GPA: 5.00 / 5.00*

Ideal School and College — 2019 – 2020
Secondary School Certificate (Science) — *GPA: 4.83 / 5.00*

Experience

Research Assistant — Jun 2025 – Present
DHMAINetRG – East West University — *Dhaka, Bangladesh*

- Conducting research on **Computer Networks and Network Protocols** under Dr. Hasan Mahmood Aminul Islam, Dept. of CSE.
- Independently authored and contributed to **literature reviews** and academic manuscripts, improving the team's publication readiness.
- Analyzed network simulation results using the **ONE Simulator**, contributing to the group's ongoing research pipeline.

Website Content Developer & QA Analyst — Oct 2025 – Jan 2026
East West University — *Dhaka, Bangladesh*

- Restructured and migrated content for a newly launched university website using a **CMS**, ensuring accurate information architecture across all pages.
- Performed systematic quality assurance across the site, **identifying and resolving content inconsistencies** before public release.

E-Commerce Seller — May 2024 – Jun 2024
Etsy — *Remote*

- Managed and maintained multiple seller accounts, consistently updating product listings to improve visibility.
- Enhanced product presentation by editing images with **Adobe Photoshop**, directly supporting sales readiness.

Publications

PRIoTP: A Partially Reliable and RL-based Context-Aware Transport Protocol for IoT Networks Driven by Edge Intelligence | IEEE
2026 22nd International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)

Integration and Validation of Map Data in the ONE Simulator: IoV in Dhaka City Perspective | IEEE
2025 28th International Conference on Computer and Information Technology (ICCIT)

The Impact of Pedestrian Movement on IoV Performance and Simulation | IEEE
2025 28th International Conference on Computer and Information Technology (ICCIT)

Design and Implementation of a Generative Pre-Trained Transformer for Universities of Bangladesh | IEEE
2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)

Projects

LapWise AI: RAG Expert Laptop Recommender | *React 19, TypeScript, FastAPI, PySpark, Qdrant* | GitHub

- Architected an end-to-end RAG expert recommendation system indexing 8,485 unique laptops across 127k passages using **BAAI/bge-base-en-v1.5** embeddings and **FAISS HNSW / Qdrant**.
- Engineered a distributed data pipeline in **Apache Spark** executing MinHash LSH deduplication and KNN missing-value imputation across multi-source scraped catalogues.
- Built an asynchronous **FastAPI** backend with hybrid metadata-vector filtering and SSE chat streaming, paired with a modern **React 19**, Vite, and Tailwind CSS dashboard.

Library Management System | *Java, JavaFX* | GitHub

- Built a **GUI-based desktop application** with role-differentiated login, enabling distinct workflows for librarians and students.
- Engineered a **file-based data persistence system**, ensuring secure and reliable storage of all records without a database dependency.

Metro Rail Ticketing System | *C* | GitHub

- Designed a fully functional CLI-based ticketing system with an intuitive, minimal-keyboard navigation interface, reducing user interaction steps.
- Implemented **role-based secure login** (user and admin), enforcing access control at the application level.

Skills

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL
Web & Frameworks: HTML, CSS, React, React Native, Node.js, Express.js, FastAPI, Vite, Tailwind CSS
Database & Big Data: MySQL, PostgreSQL, Qdrant, FAISS, Apache Spark
Tools & Platforms: Git, GitHub, GitHub Actions, Docker, VS Code, LaTeX, Adobe Photoshop, ONE Simulator
Interests: Computer Networks, Network Protocols, Retrieval-Augmented Generation (RAG), Academic Research

Extracurricular Activities

East West University Robotics Club Spring 2024 – Present

Senior Member

- Led orientation sessions for **new member cohorts**, streamlining the onboarding process for incoming students.
- Volunteered at multiple workshops and seminars, actively supporting the club's technical and community events.

East West University Cyber Security Club Fall 2024 – Present

Founding Member

- Helped **establish the club from the ground up**, contributing to its founding structure and initial membership drive.
- Organized and participated in multiple orientation programs and workshops, growing the club's active member base.